A counterexample to Greedy Sparsest Poset consistency under restricted faithfulness

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Abstract

The oracle consistency conjecture for Greedy Sparsest Poset asks whether every starting graph can reach a globally sparsest graph by legitimate mark changes and reconstruction, without increasing the edge count. We give a counterexample under its stated Markov and restricted-faithfulness assumptions. For a nondegenerate Gaussian distribution on six variables, an explicit starting poset maps to an eight-edge graph whose entire reachable component consists of three eight-edge graphs, whereas the global minimum is six. Both applications of the poset-to-graph construction and the maximal ancestral closure fix each of these three graphs. The Gaussian input is a previously published example; the obstruction is established by analyzing all possible mark changes. Exact conditional-independence and reconstruction tables are included. The example does not address consistency under full faithfulness.

1 Introduction

Ordering-based causal discovery with latent variables associates a directed maximal ancestral graph to each partial order of the observed variables. Bernstein, Saeed, Squires and Uhler [1, §§3–4] prove that global minimization of the edge count identifies the true Markov-equivalence class under restricted faithfulness. They separately conjecture that the Greedy Sparsest Poset search can reach a global minimum from every starting point [1, Conjecture 1 and Algorithm 2]. Squires and Uhler [2, §4.4, p. 1803] retain this search question.

The distinction between global minimization and reachability matters even when all conditional independences are known exactly. We use the Gaussian structural equation model in [1, Figure 1, p. 4101] and exhibit a closed three-graph component of the prescribed search. Each graph in this component has eight edges; a graph with six edges is an image of the same poset construction. Thus no search depth permits an escape from the component. The Gaussian construction is credited to [1]; the argument below concerns its search behavior. In particular, it does not challenge their global sparsest-graph theorem.

2 The search and the counterexample

A directed maximal ancestral graph (DMAG) has directed and bidirected edges, at most one edge per vertex pair, no directed cycles, and no directed path between the endpoints of a bidirected edge. On a path, an internal vertex is a collider when both incident path edges have arrowheads at that vertex. A path is *m-connecting* given a set S of other vertices when every internal noncollider is outside S and every internal collider has a directed descendant in S, allowing itself. Absence

of such a path defines m-separation. Maximality requires a separating set for every nonadjacent pair. Write $I(H)$ for the separations encoded by H; DMAGs are Markov equivalent when these separation models agree.

A distribution P is Markov to G^* if every separation in $I(G^*)$ is a conditional independence of P. Restricted faithfulness requires dependence for every m-connected pair, given every $S \subseteq V \setminus \{i, j\}$, whenever the pair is adjacent, shares a skeleton neighbor, or is the pair of endpoints of a discriminating path. A discriminating path for k has the form $\langle i, \dots, k, j\rangle$, at least three edges, nonadjacent endpoints, and every vertex strictly between i and k is both a collider on the path and a parent of j. These are the assumptions of [1, Definition 1].

For a partial order π, let

$$D_\pi(i, j) = \{v : v \leq_\pi i \text{ or } v \leq_\pi j\} \setminus \{i, j\}.$$

The graph $A(\pi, P)$ joins i, j if and only if $X_i \not\perp\!\!\!\perp X_j \mid X_{D_\pi(i,j)}$. It orients comparable endpoints from the smaller to the larger and joins incomparable endpoints by a bidirected edge. If $\mathrm{po}(H)$ is directed ancestry, including equality, the map used in the conjecture is

$$G_\pi = \mathrm{cl}_{\mathrm{ma}}(A(\mathrm{po}(A(\pi, P)), P)). \tag{1}$$

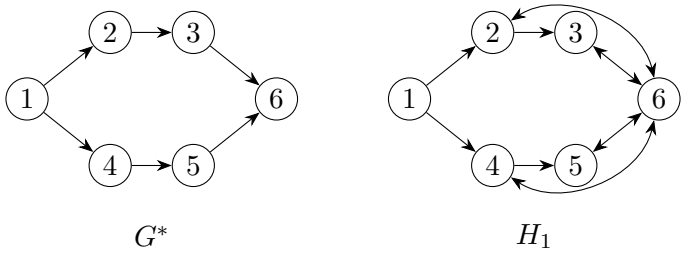
The maximal ancestral closure $\mathrm{cl}_{\mathrm{ma}}$ adds inseparable pairs and preserves the separation model and ancestral orientations. We retain both applications of A and this closure in every calculation.

The search vertices are the distinct graphs G_π. A legitimate mark change replaces one edge $a \to b$ by $a \leftrightarrow b$, or the reverse in either orientation, while retaining DMAG status and Markov equivalence. From H, the search moves to $G_{\mathrm{po}(H')}$ when H' is obtained by such a change. The conjecture asks for a finite path from every starting vertex to a global edge minimum, with nonincreasing edge counts.

Let $e_1, \dots, e_6$ be independent standard normal variables and define

$$\begin{aligned} X_1 &= e_1, & X_2 &= X_1 + e_2, & X_3 &= X_2 + e_3, \\ X_4 &= X_1 + e_4, & X_5 &= X_4 + e_5, & X_6 &= X_3 - X_5 + e_6. \end{aligned} \tag{2}$$

Denote their joint distribution by P. The six-edge true graph G^* has the two directed paths $1 \to 2 \to 3 \to 6$ and $1 \to 4 \to 5 \to 6$. Let H_1 have directed edges $1 \to 2$, $2 \to 3$, $1 \to 4$, $4 \to 5$ and bidirected edges $2 \leftrightarrow 6$, $3 \leftrightarrow 6$, $4 \leftrightarrow 6$, $5 \leftrightarrow 6$. Define H_0 by replacing $1 \to 4$ in H_1 with $1 \leftrightarrow 4$, and H_2 by replacing $1 \to 2$ in H_1 with $1 \leftrightarrow 2$.



G^* H_1

Figure 1: The six-edge graph G^* of the Gaussian model in [1, Figure 1], and the eight-edge graph H_1. Changing one of the two edges out of vertex 1 to bidirected gives H_0 or H_2.

Theorem 1. *The distribution (2) is nondegenerate Gaussian, Markov and restricted-faithful to the DMAG G^*. For the poset π_0 generated by $1 < 2 < 3$ and $4 < 5$, with all unforced pairs incomparable, $G_{\pi_0} = H_0$. Its entire reachable set in the search defined above is $\{H_0, H_1, H_2\}$, with transitions*

$$H_0 \;\rightleftarrows\; H_1 \;\rightleftarrows\; H_2. \tag{3}$$

Each graph has eight edges, whereas $\min_\pi |E(G_\pi)| = 6$. Consequently the every-start oracle consistency assertion under restricted faithfulness is false.

3 Admissibility and reconstruction

For a positive definite Gaussian covariance Σ, set

$$q_{ij}(S) = \Sigma_{ij} - \Sigma_{iS}\Sigma_{SS}^{-1}\Sigma_{Sj}, \qquad q_{ij}(\varnothing) = \Sigma_{ij}.$$

Completing the square in the joint Gaussian density gives conditional covariance $q_{ij}(S)$, so $X_i \perp\!\!\!\perp X_j \mid X_S$ if and only if $q_{ij}(S) = 0$. With the elements of S in increasing order, define

$$N_{ij}(S) = \det \Sigma_{(i,S),(j,S)} = \det(\Sigma_{SS})\, q_{ij}(S). \tag{4}$$

The second equality follows by eliminating the entries in the first row against the lower rows. The empty principal determinant is 1. All other principal determinants are positive. The same conditional-density argument for two groups of coordinates shows that zero cross-covariance entries give group conditional independence; hence pairwise comparisons suffice for the Markov property of this Gaussian distribution.

From (2), $X = Le$ with

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & -1 & -1 & 1 \end{pmatrix}, \qquad \Sigma = LL^T = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 2 & 1 & 1 & 1 \\ 1 & 2 & 3 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 2 & -1 \\ 1 & 1 & 1 & 2 & 3 & -2 \\ 0 & 1 & 2 & -1 & -2 & 5 \end{pmatrix}. \tag{5}$$

The triangular matrix L has diagonal 1, which proves positive definiteness. Table 1 lists every numerator in (4) for all 15 vertex pairs and all 16 possible conditioning sets. The entries are integer determinants of (5).

The skeleton of G^* is the cycle $1, 2, 3, 6, 5, 4, 1$. There are exactly two simple paths between any pair of distinct vertices. On either path all internal vertices other than 6 are noncolliders; vertex 6, if internal, is a collider with no proper descendants. Thus a path is connecting precisely when none of its other internal vertices lies in S and, if 6 is internal, $6 \in S$. This rule gives Table 2. Comparing the two tables shows that the conditional independences of P are exactly those encoded by G^* plus $X_1 \perp\!\!\!\perp X_6$. In particular, P is Markov. The graph is ancestral, and Table 2 provides a separator for every nonadjacent pair, proving maximality.

The only extra independence has endpoints 1 and 6 at skeleton distance three. They are neither adjacent nor endpoints of a length-two path. Endpoints of a discriminating path share its first internal vertex as a skeleton neighbor: that vertex is adjacent to the first endpoint and is a parent of the last endpoint. Therefore 1 and 6 cannot be such endpoints. All other connecting pairs, for every conditioning set, remain dependent by Table 1. This proves restricted faithfulness, with all its conditioning-set quantifiers.

We will use the following local transformation criterion.

Lemma 2 (Zhang–Spirtes, Lemma 1 of [3])**.** *Let H be a DMAG with $a \to b$. Replacing this edge by $a \leftrightarrow b$ yields a Markov-equivalent DMAG if and only if:*

1. *there is no other directed path from a to b;*
2. *every parent of a is a parent of b, and every spouse of a is a parent or spouse of b;*
3. *there is no discriminating path for a ending with a, b.*

Here a spouse is a vertex joined by a bidirected edge.

The graph H_1 is ancestral, since its directed paths stay within the rooted tree on vertices 1 through 5 and vertex 6 has no directed incident edge. For maximality, separators for its seven nonadjacent pairs are

$$(1,3) \mid \{2\}, \quad (1,5) \mid \{4\}, \quad (1,6) \mid \varnothing, \quad (i,j) \mid \{1\} \quad (i \in \{2,3\},\ j \in \{4,5\}).$$

For endpoints other than 6, every path through 6 has an unconditioned collider there; the unique path in the remaining tree is blocked by the listed separator. A simple path from 1 to 6 first follows the directed tree from 1 to some $v \in \{2,3,4,5\}$, then uses $v \leftrightarrow 6$. Its vertex v is an unconditioned collider. This proves separation of 1 and 6 as well.

Both root-edge changes $1 \to 2$ to $1 \leftrightarrow 2$ and $1 \to 4$ to $1 \leftrightarrow 4$ in H_1 satisfy Lemma 2. There is no alternative directed path; vertex 1 has neither parents nor spouses; and a discriminating path for 1 ending at $b = 2$ or 4 would require a parent of b distinct from 1, which does not exist. Thus H_0, H_1, H_2 are equivalent DMAGs.

Their ancestry posets $\pi_j = \mathrm{po}(H_j)$ are generated by

$$\begin{array}{c|l} \pi_0 & 1 < 2 < 3, \quad 4 < 5 \\ \pi_1 & 1 < 2 < 3, \quad 1 < 4 < 5 \\ \pi_2 & 2 < 3, \quad 1 < 4 < 5. \end{array} \tag{6}$$

All unforced pairs, including those involving 6, are incomparable. Table 3 gives every $D_{\pi_j}(i,j)$ and conditional covariance for $j = 0, 1$, and for $\mathrm{po}(G^*)$. Nonzero entries produce exactly the edges specified above, with their prescribed orientations. Hence $A(\pi_j, P) = H_j$ for $j = 0, 1$, and $A(\mathrm{po}(G^*), P) = G^*$. The transformation

$$(X_1, X_2, X_3, X_4, X_5, X_6) \longmapsto (X_1, X_4, X_5, X_2, X_3, -X_6)$$

preserves P: exchange (e_2, e_3) with (e_4, e_5) and replace e_6 by $-e_6$ in (2). It exchanges H_0 with H_2 and preserves conditional independence, giving $A(\pi_2, P) = H_2$. Because $\pi_j = \mathrm{po}(H_j)$, each second application of A returns the same graph. Maximality makes the closure the identity. Therefore

$$G_{\pi_j} = H_j \quad (j = 0, 1, 2), \qquad G_{\mathrm{po}(G^*)} = G^*. \tag{7}$$

Every true adjacent pair has a nonzero numerator for every conditioning set in Table 1. It therefore occurs in each application of A for every poset, and the closure cannot remove it. Consequently every G_π has at least six edges, and (7) attains this bound.

4 The closed search component

We now check every possible mark change. For a proposed change of $a \leftrightarrow b$ to $a \rightarrow b$, condition 2 of Lemma 2 is tested in the hypothetical resulting directed-edge graph. If the change were legitimate, that graph would be a DMAG and the lemma would apply in reverse. Thus an unchanged parent or spouse violating condition 2 rules the change out, without assuming the hypothetical graph is valid.

The following changes fail in every H_j; the table records an unchanged obstruction. Nonadjacency below always refers to the common skeleton.

Proposed change	Violation of condition 2
$2 \rightarrow 3$ to $2 \leftrightarrow 3$	1 is a parent of 2 in H_0, H_1, or a spouse of 2 in H_2, but is nonadjacent to 3.
$4 \rightarrow 5$ to $4 \leftrightarrow 5$	1 is a spouse of 4 in H_0, or a parent of 4 in H_1, H_2, but is nonadjacent to 5.
$2 \leftrightarrow 6$ to $2 \rightarrow 6$	1 is a parent or spouse of 2, but is nonadjacent to 6.
$3 \leftrightarrow 6$ to $3 \rightarrow 6$	$2 \rightarrow 3$ is present, but $2 \rightarrow 6$ is absent.
$4 \leftrightarrow 6$ to $4 \rightarrow 6$	1 is a parent or spouse of 4, but is nonadjacent to 6.
$5 \leftrightarrow 6$ to $5 \rightarrow 6$	$4 \rightarrow 5$ is present, but $4 \rightarrow 6$ is absent.
$2 \leftrightarrow 6$ to $6 \rightarrow 2$	4 is a spouse of 6, but is nonadjacent to 2.
$3 \leftrightarrow 6$ to $6 \rightarrow 3$	4 is a spouse of 6, but is nonadjacent to 3.
$4 \leftrightarrow 6$ to $6 \rightarrow 4$	2 is a spouse of 6, but is nonadjacent to 4.
$5 \leftrightarrow 6$ to $6 \rightarrow 5$	2 is a spouse of 6, but is nonadjacent to 5.

For the remaining two edges incident to 1, H_1 permits exactly the two changes already proved legitimate. In H_0, changing $1 \leftrightarrow 4$ to $1 \rightarrow 4$ returns H_1. Changing it to $4 \rightarrow 1$ fails because 6 is a spouse of 4 but is nonadjacent to 1. Changing $1 \rightarrow 2$ to $1 \leftrightarrow 2$ fails because 4 is a spouse of 1 but is nonadjacent to 2. The exchanged argument in H_2 permits only $1 \leftrightarrow 2$ to $1 \rightarrow 2$, returning H_1: the opposite orientation fails because 6 is a spouse of 2 nonadjacent to 1, and changing $1 \rightarrow 4$ to $1 \leftrightarrow 4$ fails because 2 is a spouse of 1 nonadjacent to 4.

There are 13 possible single mark changes in H_0, 12 in H_1, and 13 in H_2; these arguments cover them all, including both orientations of each bidirected edge. The only legitimate changes are $H_0 \rightarrow H_1$, $H_1 \rightarrow H_0$, $H_1 \rightarrow H_2$, and $H_2 \rightarrow H_1$. Reconstruction fixes every changed graph by (7), proving (3).

Completion of the proof of Theorem 1. Section 3 establishes all distributional assumptions and shows that the specified starting poset maps to H_0. Every search path from H_0 remains among the three graphs just analyzed. Every visited graph has eight edges, while the global minimum is six. No path reaches a minimum, even if increasing moves were allowed, and in particular no nonincreasing path does so. □

The component also misses the true Markov-equivalence class directly: G^* separates 2 and 6 given $\{1, 3\}$, whereas these vertices are adjacent in every H_j. The obstruction uses the marginal cancellation $X_1 \perp\!\!\!\perp X_6$, which full faithfulness would exclude. We make no conclusion about full faithfulness, generic coefficients, alternative search moves, or finite-sample inference.

A Exact conditional-independence tables

For each pair $i < j$, let $s_1 < s_2 < s_3 < s_4$ be the other vertices. Encode S by $c(S) = \sum_{k:s_k \in S} 2^{k-1}$, between 0 and 15. Table 1 gives $N_{ij}(S)$, and Table 2 gives the codes of separating sets in G^*. Thus these tables can be checked directly from (5) and the two-path rule in Section 3; no enumeration of all posets is needed. The optional accompanying Python checker uses rational arithmetic and

exhaustive simple-path separation tests to reproduce the tables, the four fixed points, and all legitimate transitions.

(i,j)	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$(1,2)$	1	1	1	1	2	2	1	1	5	3	3	2	4	2	2	1
$(1,3)$	1	0	1	0	2	0	1	0	5	1	2	0	2	0	1	0
$(1,4)$	1	1	2	1	1	1	2	1	5	3	4	2	3	2	2	1
$(1,5)$	1	1	2	1	0	0	0	0	5	2	2	1	1	0	0	0
$(1,6)$	0	-1	-2	-1	1	0	0	0	2	0	0	0	1	0	0	0
$(2,3)$	2	1	3	1	5	2	3	1	8	3	6	2	5	2	3	1
$(2,4)$	1	0	1	0	1	0	1	0	6	1	2	0	2	0	1	0
$(2,5)$	1	0	1	0	0	0	0	0	7	2	1	0	3	1	0	0
$(2,6)$	1	1	-1	0	3	1	0	0	5	2	0	0	3	1	0	0
$(3,4)$	1	0	0	0	1	0	0	0	7	2	3	1	1	0	0	0
$(3,5)$	1	0	0	0	0	0	0	0	9	4	5	2	5	2	3	1
$(3,6)$	2	2	2	1	5	2	3	1	8	4	5	2	5	2	3	1
$(4,5)$	2	1	3	1	5	2	3	1	8	3	6	2	5	2	3	1
$(4,6)$	-1	-1	-3	-1	-5	-2	-3	-1	1	0	0	0	0	0	0	0
$(5,6)$	-2	-2	-5	-2	-8	-4	-5	-2	-2	-1	-3	-1	-5	-2	-3	-1

Table 1: All 240 almost-principal determinant numerators $N_{ij}(S)$. The column is the conditioning-set code $c(S)$ defined above.

(i,j)	Other vertices in bit order	m-separating codes in G^*
$(1,2)$	$3,4,5,6$	$\varnothing$
$(1,3)$	$2,4,5,6$	$1,3,5,7,11,13,15$
$(1,4)$	$2,3,5,6$	$\varnothing$
$(1,5)$	$2,3,4,6$	$4,5,6,7,13,14,15$
$(1,6)$	$2,3,4,5$	$5,6,7,9,10,11,13,14,15$
$(2,3)$	$1,4,5,6$	$\varnothing$
$(2,4)$	$1,3,5,6$	$1,3,5,7,11,13,15$
$(2,5)$	$1,3,4,6$	$1,3,4,5,6,7,11,14,15$
$(2,6)$	$1,3,4,5$	$3,6,7,10,11,14,15$
$(3,4)$	$1,2,5,6$	$1,2,3,5,6,7,13,14,15$
$(3,5)$	$1,2,4,6$	$1,2,3,4,5,6,7$
$(3,6)$	$1,2,4,5$	$\varnothing$
$(4,5)$	$1,2,3,6$	$\varnothing$
$(4,6)$	$1,2,3,5$	$9,10,11,12,13,14,15$
$(5,6)$	$1,2,3,4$	$\varnothing$

Table 2: Separations in the true graph. Empty code lists mean there is no separating set, rather than that the empty set separates. Comparing with Table 1 leaves only the extra independence $(1,6 \mid \varnothing)$.

B Reconstruction tables

Table 3 lists all conditioning sets $D = D_{\mathrm{po}(H)}(i,j)$ and values $q = q_{ij}(D)$ needed for the first application of A. Each resulting graph has the same ancestry poset and is maximal, which also checks the second application and closure in (1). The symmetry in Section 3 gives the H_2 table.

	H_0		H_1		G^*	
(i,j)	D	q	D	q	D	q
$(1,2)$	$\varnothing$	1	$\varnothing$	1	$\varnothing$	1
$(1,3)$	$\{2\}$	0	$\{2\}$	0	$\{2\}$	0
$(1,4)$	$\varnothing$	1	$\varnothing$	1	$\varnothing$	1
$(1,5)$	$\{4\}$	0	$\{4\}$	0	$\{4\}$	0
$(1,6)$	$\varnothing$	0	$\varnothing$	0	$\{2,3,4,5\}$	0
$(2,3)$	$\{1\}$	1	$\{1\}$	1	$\{1\}$	1
$(2,4)$	$\{1\}$	0	$\{1\}$	0	$\{1\}$	0
$(2,5)$	$\{1,4\}$	0	$\{1,4\}$	0	$\{1,4\}$	0
$(2,6)$	$\{1\}$	1	$\{1\}$	1	$\{1,3,4,5\}$	0
$(3,4)$	$\{1,2\}$	0	$\{1,2\}$	0	$\{1,2\}$	0
$(3,5)$	$\{1,2,4\}$	0	$\{1,2,4\}$	0	$\{1,2,4\}$	0
$(3,6)$	$\{1,2\}$	1	$\{1,2\}$	1	$\{1,2,4,5\}$	1
$(4,5)$	$\varnothing$	2	$\{1\}$	1	$\{1\}$	1
$(4,6)$	$\varnothing$	-1	$\{1\}$	-1	$\{1,2,3,5\}$	0
$(5,6)$	$\{4\}$	-1	$\{1,4\}$	-1	$\{1,2,3,4\}$	-1

Table 3: Exact adjacency calculations for H_0, H_1, and G^*. An edge occurs precisely when $q \neq 0$, with orientation determined by the poset.

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[3] J. Zhang and P. Spirtes. A transformational characterization of Markov equivalence for directed acyclic graphs with latent variables. In *Proceedings of the Twenty-First Conference on Uncertainty in Artificial Intelligence*, 2005, §2.2, Lemma 1. https://arxiv.org/abs/1207.1419v1.

AI use. OpenAI Codex assisted with the search for the obstruction, exact-arithmetic code, proof development, source checking, and manuscript preparation. AI assistance does not constitute human or formal verification. Independent mathematical review is requested.